AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *[GEN26-A0H-I004]*

Team Name: *[ZENITH]*

Domain: *[Intelligent Systems Digital Futures]*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

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Presentation Outline

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Problem Statement

[Our team aims to solve the problem of high cloud dependency, slow processing, and privacy risks in AI applications. Currently, many AI systems rely on centralized cloud servers, which require stable internet connectivity and may expose sensitive data during transmission. Coordinating multiple AI agents across edge devices with limited memory and processing power is also challenging. Our goal is to develop a lightweight workflow engine that enables AI agents to coordinate and process tasks locally, reducing cloud dependency, improving data privacy, and enabling faster, more reliable AI processing.]

Problem Understanding

- [This problem affects AI developers, hospitals, industries, and IoT users who face processing delays, privacy risks, and internet connectivity issues. Our solution enables AI agents to work together locally, reducing cloud dependency and improving speed, privacy, and reliability.]
- [The problem is caused by heavy dependence on centralized cloud servers, unstable internet connectivity, delays in data transmission, privacy concerns, and limited memory and processing power on edge devices. Coordinating multiple AI agents efficiently across distributed devices is also challenging.]
- [It causes slower responses, higher cloud costs, privacy risks, and service interruptions when internet connectivity is poor. As a result, hospitals, industries, AI developers, and IoT users may experience reduced efficiency, reliability, and performance.]

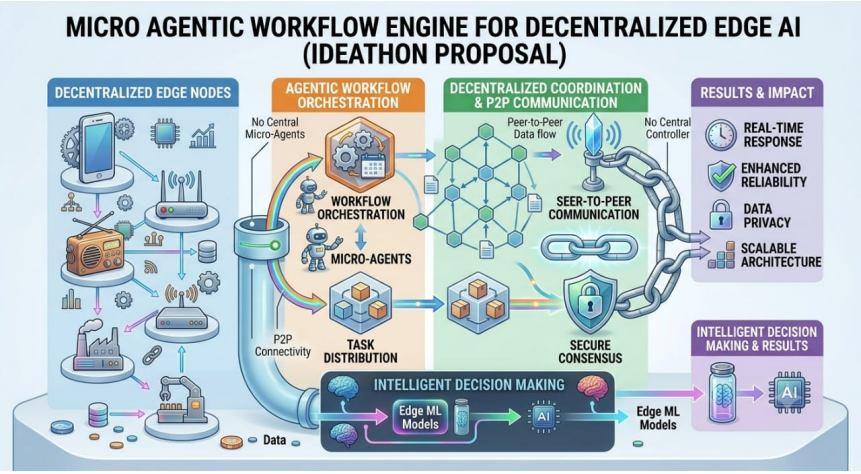
Existing Gaps & Challenges

- [Existing Gap or Limitation 1: Many existing AI systems depend heavily on cloud servers, causing delays, connectivity issues, and privacy concerns when processing data.]
- [Existing Gap or Limitation 2: Existing edge devices have limited memory and processing power, making it difficult to coordinate multiple AI agents efficiently and reliably.]
- [Main Challenge That Remains Unsolved: Developing a lightweight and reliable workflow engine that coordinates multiple AI agents across resource-limited edge devices while maintaining data privacy, efficient resource usage, and smooth operation even with poor internet connectivity.]

Our Proposed Solution

- [Core Idea or Product: A lightweight workflow engine that enables multiple AI agents to coordinate, share tasks, and process data locally across edge devices, reducing cloud dependency while improving speed, privacy, and efficiency.]
- [Our solution works by breaking a task into smaller jobs and assigning them to different AI agents on nearby edge devices. The workflow engine coordinates these agents, collects their results, and produces a final output locally whenever possible, reducing cloud dependency, delays, and unnecessary data sharing.]
- [Main Value Delivered to Users: Faster AI processing, improved data privacy, reduced cloud costs, and reliable task execution even with limited internet connectivity.]

Technical Approach



[The workflow engine receives data, divides tasks among multiple AI agents on edge devices, and coordinates their execution. Each agent processes its assigned task, and the engine combines their results to produce a final output locally, reducing cloud dependency, improving privacy, and enabling faster responses.]

Tech Stack

Frontend:[User Dashboard] **Database:** [Data Storage]

Backend:[Workflow Engine]

Innovation & Uniqueness

- [Innovation: Lightweight, decentralized AI agents that coordinate tasks locally with reduced cloud dependency, improved privacy, and faster responses.]
- [This approach enables local processing, reduces cloud dependency and delays, protects sensitive data, and efficiently coordinates AI tasks even on devices with limited resources.]
- [Key Differentiator: Lightweight, decentralized AI agent coordination with local processing and reduced cloud dependency.]

Feasibility & Scalability

- **Feasibility:** [Resources: Python, lightweight AI models, edge devices, and a workflow engine.
  Steps: Design, develop, integrate, test, and optimize the prototype.]
- **Scalability:** [Add more edge devices and AI agents to handle larger workloads across multiple locations.]
- **Risks:** [Constraints: Limited memory, processing power, connectivity, and security risks.
  Mitigations: Lightweight models, efficient task scheduling, offline processing, and secure communication.]

Cost for Product (Optional)

- Development or hardware cost: [Development Cost: Low to moderate, using open-source tools and existing devices.
  Hardware Cost: Minimal, using laptops or affordable edge devices.]
- Hosting or maintenance cost: [Hosting Cost: Low, with local processing and minimal cloud usage.
  Maintenance Cost: Moderate, for software updates, security, and performance monitoring.]
- Estimated total: [Estimated Total Cost: 5,000–15,000 rupees for a basic prototype using existing laptops and open-source tools.]

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: [Hospitals — Process patient sensor data locally to detect abnormal readings and generate quick alerts.]
- Use case 2: [Smart Factories — Process sensor data locally to detect machine faults and trigger quick alerts.]
- Use case 3: [Smart Agriculture — Analyze soil and crop sensor data locally to support irrigation decisions and detect unusual conditions.]
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- [Benefits: Faster processing, improved privacy, reduced cloud costs, and reliable operation for hospitals, industries, and farmers.]
- [Social: Improved data privacy and access to reliable AI. Economic: Reduced cloud and operational costs. Environmental: Lower network usage and potentially reduced energy consumption.]
- [Expected Measurable Outcomes: Reduced processing latency, lower cloud data usage, improved task success rate, and better resource efficiency.]

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG [9]	Target [9.5]	[Industry, Innovation and Infrastructure]	[Promotes efficient, innovative, and decentralized AI technology.]
SDG [12]	Target [12.2]	[Responsible Consumption and Production]	[Reduces unnecessary cloud usage and encourages efficient use of computing resources.

Individual Member Contribution

Member	Contribution or responsibility
[SARUMATHI R N]	[Team coordination, planning, and overall project management.]
[KRISHNA MANOJ JADHAO]	[Research, problem analysis, and solution design.]
[SABITHA J]	[Frontend design and user interface development.]
[KAVYA NAGAVARDINI N]	[Backend workflow engine, database, and testing.]

Yukti ID

[ir20266405]